

Гадиров Наиль
задание 18_1

1) создаем файл и заполняем

```
vexpress-v2p-ca9.dtb  zImage
gadirov@nail-gad:~/Desktop/HW_18$ touch init.c
gadirov@nail-gad:~/Desktop/HW_18$ vim init.c
gadirov@nail-gad:~/Desktop/HW_18$ cat init.c
#include <stdio.h>
#include <unistd.h>
int main(void)
{
    printf("====HELLO====");
    sleep(20);
}
gadirov@nail-gad:~/Desktop/HW_18$
```

2) компилируем

```
arm-linux-gnueabihf-gcov-13      arptables-restore
arm-linux-gnueabihf-gcov-dump    arptables-save
gadirov@nail-gad:~/Desktop/HW_18$ arm-linux-gnueabihf-gcc -static init.c -o init
gadirov@nail-gad:~/Desktop/HW_18$ ls
init  init.c  vexpress-v2p-ca9.dtb  zImage
gadirov@nail-gad:~/Desktop/HW_18$
```

3) создаем корневой файл

```
init  init.c  vexpress-v2p-ca9.dtb  zImage
gadirov@nail-gad:~/Desktop/HW_18$ echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
829 блоков
gadirov@nail-gad:~/Desktop/HW_18$ ls
init  init.c  initramfs.cpio.gz  vexpress-v2p-ca9.dtb  zImage
gadirov@nail-gad:~/Desktop/HW_18$
```

4) прошиваем

```
motherboard-bus@400000000/bus@400000000:motherboard-bus@400000000:iofpga@7,00000000/10006000.k
mi/serio0/input/input0
[ 1.233178] Loading compiled-in X.509 certificates
[ 1.244870] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a P
L011 rev1
[ 1.246618] printk: console [ttyAMA0] enabled
[ 1.248234] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a P
L011 rev1
[ 1.249152] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a P
L011 rev1
[ 1.275686] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a P
L011 rev1
[ 1.283043] clk: Disabling unused clocks
[ 1.283330] PM: genpd: Disabling unused power domains
[ 1.314084] Freeing unused kernel image (initmem) memory: 2048K
[ 1.317503] Run /init as init process
[ 1.641001] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@400000000/bus
@400000000:motherboard-bus@400000000/bus@400000000:motherboard-bus@400000000:iofpga@7,00000000/
10007000.kmi/serio1/input/input2
[ 11.360207] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 11.360300] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 11.360323] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 11.360339] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
=====HELLO=====
[ 21.366827] Kernel panic - not syncing: Attempted to kill init!
exitcode=0x00000000
[ 21.367403] CPU: 0 UID: 0 PID: 1 Comm: init Tainted: G W 7.1.0-07602-g6
6affa37cfac #1 VOLUNTARY
[ 21.367519] Tainted: [W]=WARN
[ 21.367555] Hardware name: ARM-Versatile Express
[ 21.367612] Call trace:
[ 21.367643] unwind_backtrace from show_stack+0x10/0x14
[ 21.367773] show_stack from dump_stack_lvl+0x54/0x68
```

задание 18_2

1) делаем дефолтный конфиг

```
Linux kernel printk buffer support (FEATURE_KMSG_SYSLOG) [Y/n/?] (NEW) y
dirov@nail-gad:~/prog/kernel/FS/busybox-1.36.1$ ARCH=arm make menuconfig
HOSTLD scripts/kconfig/mconf
HOSTCC scripts/kconfig/lxdialog/checklist.o
HOSTCC scripts/kconfig/lxdialog/inputbox.o
HOSTCC scripts/kconfig/lxdialog/lxdialog.o
HOSTCC scripts/kconfig/lxdialog/menubox.o
HOSTCC scripts/kconfig/lxdialog/msgbox.o
HOSTCC scripts/kconfig/lxdialog/textbox.o
HOSTCC scripts/kconfig/lxdialog/util.o
HOSTCC scripts/kconfig/lxdialog/yesno.o
HOSTLD scripts/kconfig/lxdialog/lxdialog
ripts/kconfig/mconf Config.in

using defaults found in .config
```

2) делаем статиком и добавляем префикс

```
[ ] Support LOG_INFO level syslog messages
--- Build Options
[*] Build static binary (no shared libs)
[ ] Force NOMMU build
() Cross compiler prefix
() Path to sysroot
() Additional CFLAGS
() Additional LDFLAGS
() Additional LDLIBS
```

```
[ ] exec prefers applets
(/proc/self/exe) Path to busybox executable
[ ] Support NSA Security Enhanced Linux
[ ] Clean up all memory before exiting (usually not needed)
[*] Support LOG_INFO level syslog messages
--- Build Options
[*] Build static binary (no shared libs)
[ ] Force NOMMU build
(arm-linux-gnueabihf-) Cross compiler prefix
() Path to sysroot
() Additional CFLAGS
() Additional LDFLAGS
() Additional LDLIBS
[ ] Avoid using GCC-specific code constructs
[*] Use -mpreferred-stack-boundary=2 on i386 arch
[*] Use -static-libgcc
Test-11: test-11: Options ("test-11" behavior)
```

3)
создаем и делаем install

```
gadirov@nail-gad:~/prog/kernel/FS/busybox-1.36.1$ make -j16
SPLIT include/autoconf.h -> include/config/*
GEN include/bbconfigopts.h
GEN include/common_bufsiz.h
GEN include/embedded_scripts.h
HOSTCC applets/usage
HOSTCC applets/applet_tables
applets/usage.c: In function 'main':
applets/usage.c:52:17: warning: ignoring return value of 'write' declared with attribute 'warn_unused_result' [-Wunused-result]
   52 |                 write(STDOUT_FILENO, usage_array[i].usage, strlen(usage) + 1);
      |                 ^
GEN include/usage_compressed.h
```

4) записываем в initramfs.cpio.gz и переносим

```
drwxrwxr-x  4 gadirov gadirov 4096 Jun 18 06:03 usr
gadirov@nail-gad:~/prog/kernel/FS/busybox-1.36.1/_install$ find . | cpio -o -H newc | gzip >
initramfs.cpio.gz
cpio: Файл ./initramfs.cpio.gz увеличился, 786432 новых байта не скопированы
5157 блоков
gadirov@nail-gad:~/prog/kernel/FS/busybox-1.36.1/_install$ cd ..
```

5) запускаем гету с нашей корневой файловой системой

```
Library resolv is needed, can't exclude it (yet)
Final link with: m resolv
gadirov@nail-gad:~/prog/kernel/FS/busybox-1.36.1$ file busybox
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, Bu
ildID[sha1]=ef53a747e4ee17f08174e70f108dc5a0258157f6, for GNU/Linux 3.2.0, stripped
gadirov@nail-gad:~/prog/kernel/FS/busybox-1.36.1$
```

6) результат и добавление прос для работы ps

```
1.418247] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a P
011 rev1
1.420321] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a P
011 rev1
1.425375] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a P
011 rev1
1.441599] clk: Disabling unused clocks
1.441834] PM: genpd: Disabling unused power domains
1.481889] Freeing unused kernel image (initmem) memory: 2048K
1.484860] Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
- # [ 1.838195] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000
/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000
000/10007000.kmi/serio1/input/input2
11.997795] amba 1000f000.wdt: deferred probe pending: (reason unknown)
11.997871] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
11.997884] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
11.997894] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
ls
bin          initramfs.cpio.gz  root          usr
dev          linuxrc           sbin
- # ip a
1: lo: <LOOPBACK> mtu 65536 qdisc noop qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000
    link/ether 52:54:00:12:34:56 brd ff:ff:ff:ff:ff:ff
- #
```



```
38 0      0:00 [kworker/u17:1-p]
39 0      0:00 [oom_reaper]
40 0      0:00 [kworker/R-write]
41 0      0:00 [kcompactd0]
42 0      0:00 [kworker/R-kbloc]
43 0      0:00 [kworker/u17:2-p]
44 0      0:00 [kworker/R-ata_s]
45 0      0:00 [kworker/R-edac-]
46 0      0:00 [kworker/R-devfr]
47 0      0:00 [watchdogd]
48 0      0:00 [kworker/R-rpcio]
49 0      0:00 [kworker/R-xprti]
50 0      0:00 [kworker/u17:3-a]
51 0      0:00 [kswapd0]
52 0      0:00 [kworker/R-nfsio]
53 0      0:00 [kworker/u18:0]
54 0      0:00 [kworker/u17:4-a]
55 0      0:00 [kworker/u17:5-a]
56 0      0:00 [kworker/u17:6-p]
57 0      0:00 [kworker/u17:7-p]
58 0      0:00 [kworker/u17:8]
62 0      0:00 [irq/31-mmci-pl1]
63 0      0:00 [kworker/0:2-rcu]
64 0      0:00 [kworker/R-mld]
65 0      0:00 [kworker/R-ipv6_]
66 0      0:00 [kworker/u16:1-i]
67 0      0:00 [pr/ttyAMA-1]
70 0      0:00 [kworker/0:3-rcu]
78 0      0:00 ps
#
```